Guillermo Terrén-Serrano

Environmental Markets Lab (emLab), University of California, Santa Barbara

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Google Scholar

I am a computational power systems researcher working on the planning and operation of low-carbon electricity systems under uncertainty using optimization and probabilistic forecasting methods. My research integrates power system modeling, machine learning, energy meteorology, and electricity market analysis to study grid reliability, resource adequacy, and infrastructure investment under evolving climate and socioeconomic conditions. As an educator and mentor, I am committed to building inclusive research environments where students develop technical skills, scientific curiosity, and the ability to solve problems collaboratively.

Research Interests — *Power systems; machine learning; stochastic optimization; resource adequacy and reliability; probabilistic forecasting; electricity markets; low-carbon electricity systems; uncertainty quantification; energy meteorology and remote sensing.*

Education

Ph.D. in Electrical Engineering (Pass with Distinction) 2016–2022
"Intra-hour Solar Forecasting using Cloud Dynamics Features Extracted from Ground-Based Infrared Sky Images." Department of Electrical & Computer Engineering, University of New Mexico, NM, USA. *Advisor: Prof. Martínez-Ramón*

M.Sc. in Power & Energy 2015–2016
"Machine Learning Approach for Global Solar Radiation Time-Series Forecasting." Department of Electrical & Computer Engineering, University of New Mexico, NM, USA. *Advisor: Prof. Martínez-Ramón*

B.Sc. in Electrical Engineering 2006–2011
Escuela de Ingeniería y Arquitectura, Universidad de Zaragoza, Zaragoza, Spain. *Semester abroad: Pontifícia Universidade Católica do Paraná, PR, Brazil.* *Advisor: Prof. Ibáñez-Carabantes*

Professional Experience

Postdoctoral Scholar 2022–Present
Environmental Markets Lab (emLab) and Environmental Studies Program, University of California, Santa Barbara, CA, USA.

Postdoctoral Research Associate 2022–2023
Department of Statistics and Applied Probability, University of California, Santa Barbara, CA, USA.

Graduate Research Assistant 2015–2022
Center for Emerging Energy Technologies (CEET) and Department of Electrical & Computer Engineering, University of New Mexico, NM, USA.

Project Engineer 2012–2013
Photovoltaic power plants, Enerland Group, Zaragoza, Spain.

Undergraduate Research Assistant 2012
Electronic Engineering Department, Institute of Technology for Development (LACTEC), PR, Brazil.

Peer-Reviewed Publications

(†indicates equal contribution) citations=540, h-index=11, i10-index=11

13. **G. Terrén-Serrano**, R. Deshmukh and M. Martínez-Ramón. Probabilistic Day-Ahead Forecasting of System-Level Renewable Energy and Electricity Demand. *Nature Communications*, 2026.
12. **G. Terrén-Serrano** and M. Ludkovski. Extreme Day-Ahead Renewables Scenario Selection in Power Grid Operations. *Applied Energy*, 2025.
11. Q. Paletta, **G. Terrén-Serrano**, Y. Nie, B. Li, J. Bieker, W. Zhang, L. Dubus, S. Dev and C. Feng. Advances in Solar Forecasting: Computer Vision with Deep Learning. *Advances in Applied Energy*, 2023.
10. **G. Terrén-Serrano** and M. Martínez-Ramón. Deep Learning for Intra-Hour Solar Forecasting with Fusion of Features Extracted from Infrared Sky Images. *Information Fusion*, 2023.
9. **G. Terrén-Serrano** and M. Martínez-Ramón. Detection of clouds in multiple wind velocity fields using ground-based infrared sky images. *Knowledge-Based Systems*, 2023.
8. **G. Terrén-Serrano** and M. Martínez-Ramón. Kernel Learning for Intra-Hour Solar Forecasting with Infrared Sky Images and Cloud Dynamic Feature Extraction. *Renewable and Sustainable Energy Reviews*, 2023.
7. **G. Terrén-Serrano** and M. Martínez-Ramón. Processing of Global Solar Irradiance and Ground-Based Infrared Sky Images for Solar Nowcasting and Intra-Hour Forecasting Applications. *Solar Energy*, 2023.
6. **G. Terrén-Serrano** and M. Martínez-Ramón. Geospatial Perspective Reprojections for Ground-Based Sky Imaging Systems. *IEEE Transactions on Geoscience and Remote Sensing*, 2022.
5. **G. Terrén-Serrano** and M. Martínez-Ramón. Comparative Analysis of Methods for Cloud Segmentation in Ground-Based Infrared Images. *Renewable Energy*, 2021.
4. **G. Terrén-Serrano** and M. Martínez-Ramón. Multi-Layer Wind Velocity Field Visualization in Infrared Images of Clouds for Solar Irradiance Forecasting. *Applied Energy*, 2021.
3. **G. Terrén-Serrano**, A. Bashir, T. Estrada and M. Martínez-Ramón. Girasol, a Sky Imaging and Global Solar Irradiance Dataset. *Data in Brief*, 2021.
2. A. Mammoli, **G. Terrén-Serrano**, A. Menicucci, T.P. Caudell and M. Martínez-Ramón. An Experimental Method to Merge Far-Field Images from Multiple Long-Wave Infrared Sensors for Short-Term Solar Forecasting. *Solar Energy*, 2019.
1. O. García-Hinde, **G. Terrén-Serrano**, M.Á. Hombrados-Herrera, V. Gómez-Verdejo, S. Jiménez-Fernández, C. Casanova-Mateo, J. Sanz-Justo, M. Martínez-Ramón and S. Salcedo-Sanz. Evaluation of Dimensionality Reduction Methods Applied to Numerical Weather Models for Solar Radiation Forecasting. *Engineering Applications of Artificial Intelligence*, 2018.

Manuscripts Under Review

G. Terrén-Serrano and M. Ludkovski. Nearest Neighbor Functional Curve Completion for Intraday Probabilistic Revisions of Renewable Generation Forecasts. *Submitted* (preprint on arXiv), 2026.

G. Terrén-Serrano†, R. Deshmukh†, S. Dukkipati, M. Meng, S. Prasad, A. Mileva and A. Gambhir. A multi-model framework for identifying affordable low-carbon electricity pathways for India under uncertainty. *Submitted* (preprint on arXiv), 2026.

Manuscripts in Preparation

G. Terrén-Serrano, A. Sharma, R. Deshmukh and S. Prasad. Ensuring Resource Adequacy in Power Systems under Rapid Demand Growth and Decarbonization. *In preparation.*

G. Terrén-Serrano and R. Deshmukh. The Reliability Value of Variable Renewable Energy Resources in Large Decarbonizing Power Grids. *In preparation.*

S. Prasad, **G. Terrén-Serrano**, R. Deshmukh, and W. T. Wang, . Employment, Health, and Environmental Impacts of India's Power System Low-Carbon Transition. *In preparation.*

Conference Proceedings

G. Terrén-Serrano, R. Deshmukh and M. Martínez-Ramón. Day-Ahead Operational Forecast of Aggregated Solar Generation Assimilating Mesoscale Meteorology Information. *IEEE PES Grid Edge Technologies Conference & Exposition*, San Diego, USA, 2025.

G. Terrén-Serrano and M. Martínez-Ramón. Wind Flow Estimation in Thermal Sky Images for Sun Occlusion Prediction. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Segmentation Algorithms for Ground-Based Infrared Cloud Images. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022.

G. Terrén-Serrano and M. Martínez-Ramón. Explicit Basis Function Kernel Methods for Cloud Segmentation in Infrared Images. *Energy Reports*, 2022.

Grants and Funding

(*indicates co-PI-equivalent; UC postdoctoral scholars at are not eligible to serve as PI or co-PI)

Resource Adequacy and Long-Term Planning of India's Electricity System. *World Resources Institute (WRI)*, 2024–2026. PI: R. Deshmukh. Role: Investigator. Total: $450,000.

CNSI Climate Innovation Fund. *California NanoSystems Institute*, 2023. Computing resources and conference travel. PI: R. Deshmukh. Role: Co-PI*. Total: $30,000.

Research Seed Program. *UCSB Institute for Energy Efficiency*, 2022. Software development and conference travel. PI: R. Deshmukh. Role: Co-PI*. Total: $50,000.

Stochastic Models, Indices & Optimization Algorithms for Pricing & Hedging Reliability Risks in Modern Power Grids. *ARPA-E PERFORM*, 2022–2023. PI: R. Carmona. Role: Investigator. Total: $3,500,000.

The New Mexico SMART Grid Center: Sustainable, Modular, Adaptive, Resilient, and Transactive. *NSF EPSCoR*, 2018–2025. PI: G. Balakrishnan. Role: Investigator. Total: $20,000,000.

Honors and Awards

2023–2025 CNSI Climate Innovation Postdoctoral Fellowship. *California NanoSystems Institute.* Competitive fellowship to research technologies for California's climate goals. Supported extending the day-ahead solar generation forecasting framework (2022 Research Seed Program) to wind and electricity demand.

2015–2022 UNM-Avangrid Graduate Scholarship. *Avangrid (Iberdrola) Foundation.* Competitive scholarship to complete doctoral studies in the ECE department at UNM under the King Felipe VI of Spain Endowed Chair.

2011 UZ-Santander Americampus Scholarship. *Santander Bank.* Competitive scholarship to complete senior project design on an academic exchange year at the PUC-PR, Curitiba, Brazil.

Teaching and Mentoring Experience

Instructor of record:

ES 105 Renewable Energy Systems. Environmental Studies Program, UCSB. Spring 2024.

Teaching assistant:

ECE 213 Circuit Analysis II. Fall 2015, Spring 2016, Electrical & Computer Engineering (ECE), UNM.

ECE 203 Circuit Analysis I. Fall 2016, Spring 2017, ECE, UNM.

ENG 302 Fundamentals of Engineering: Electronic Circuits. Spring 2017, School of Engineering, UNM.

ECE 314 Signals and Systems. Spring 2015, ECE, UNM.

ECE 331 Data Structures and Algorithms. Spring 2015, ECE, UNM.

Guest lecturer:

ECE 417 Advanced Machine Learning. Probabilistic Principal Component Analysis. Fall 2017, ECE, UNM.

Mentor:

Bryan Lu. Research project, CETlab, Summer 2026, UCSB.

Zekai (Zach) Wang. Research project, CETlab, 2026, UCSB.

Steven De Jesus. EUREKA! Program, Summer 2023, UCSB.

David Lopez. EUREKA! Program + CAMP Research School Year Internship, 2023, UCSB.

Alexander Santos-Lozada. CIESESE Program, Summers 2017 and 2018, UNM.

Invited Talks and Conference Presentations

2026 Macro-Energy Systems (MES) Workshop. *Poster.* Georgia Tech, GA, USA.

2025 Forecasting & Markets Workshop. *Presentation.* ESIG, TN, USA.

2025 Institute for Energy Efficiency (IEE). *Seminar.* UCSB, CA, USA.

2024 ARPA-E Energy Innovation Summit. *Exhibition Booth.* Department of Energy, TX, USA.

2024 MES Workshop. *Flash Talk + Poster.* Princeton University, NJ, USA.

2024 MIT Energy Initiative. *Seminar.* Massachusetts Institute of Technology, MA, USA.

2023 International Energy Workshop (IEW). *Presentation.* Colorado School of Mines, CO, USA.

Datasets, Models, and Packages

G. Terrén-Serrano, R. Deshmukh, S. Dukkipati, M. Meng, S. Prasad, A. Mileva and A. Gambhir. *GridPath-India long-term (2020–2050) power system planning model data.* Dryad, (2026). Dataset & Model.

G. Terrén-Serrano, S. Prasad and R. Deshmukh. *GridPath-India workshops: illustrative models for Rajasthan, Madhya Pradesh, and Tamil Nadu.* GitHub, (2025). Model.

G. Terrén-Serrano, A. Bashir, T. Estrada and M. Martínez-Ramón. *Girasol, a sky imaging and global solar irradiance dataset.* Dryad, (2021). Dataset.

Institutional Service and Community Involvement

Reviewer. Applied Energy; Renewable Energy; Knowledge-Based Systems; Solar Energy; Data in Brief; IEEE Journal of Emerging and Selected Topics in Industrial Electronics; Wiley (book proposal).

Spanish Translator. Parent-teacher conferences, South Valley Academy, NM, USA (2020).

Skills

Programming Languages: Python, R, Matlab, C/C++, Bash, LabVIEW, Simulink, HTML, Pascal.

Scientific Computing: CPU, GPU, and MPI parallel computing.

Prototyping: Arduino and Raspberry Pi, and 3D printing (CAD).

Modules: TensorFlow, Keras, PyTorch, GridPath, PyPSA.

OS: Linux, Ubuntu, macOS, Windows, Raspbian.

Writing: LaTeX, Overleaf, Microsoft Office, Google Docs.

Version Control: Git, GitHub.

Languages: Spanish (native), Portuguese (fluent).

Professional References

Prof. Ranjit Deshmukh (rdeshmukh@ucsb.edu). *Associate Professor.* Bren School of Environmental Science & Management, and Environmental Studies Program, UCSB, CA, USA.

Prof. Manel Martínez-Ramón (manel@unm.edu). *Professor.* Department of Electrical and Computer Engineering, UNM, NM, USA.

Prof. Michael Ludkovski (mludkov@ucsb.edu). *Professor.* Department of Statistics and Applied Probability, UCSB, CA, USA.

Dr. Andrea Mammoli (aamammo@sandia.gov). *Principal Member of Technical Staff.* Sandia National Laboratories, NM, USA.